



Braess's paradox

COMPUTER SCIENCE AND MATHEMATICS PROJECT

THEME: TRANSPORTATION

GITHUB : [HTTPS://GITHUB.COM/KORATY/BRAESS PARADOX](https://github.com/KORATY/BRAESS_PARADOX)

SUMMARY

I. INTRODUCTION

- I.A: Case study
- I.B: Theoretical example

II. PRESENTATION OF THE ALGORITHM

- II.A: Initialization Process
- II.B: Roads (Arcs)
- II.C: Blocks
- II.D: The Interface

III. ADVICE FROM THE CONTACT

- III.A: Advice
- III.B: Curve proposals

IV. TRAFFIC ANALYSIS

- IV.A: Temporal distribution
- IV.B: 6 cities
- IV.C: 15 cities
- IV.D: The Price of Anarchy

V. STUDY OF THE PARADOX

- V.A: On the road closed and its surrounding
- V.B: For everyone

VI. CONCLUSION

VII. ANNEX

- VII.A: Iteration Process
- VII.B: Road capacity
- VII.C: The Law of Speed
- VII.D: Presentation of the contact
- VII.E Effect of social routing on a large workforce

I. INTRODUCTION

I.A: Case study

- ▶ In 1990, on the occasion of "Earth Day", the municipality of New York decided to close 42nd Street to traffic.
- ▶ Impact: Traffic has become more fluid!



I. INTRODUCTION

I.B: Theoretical example

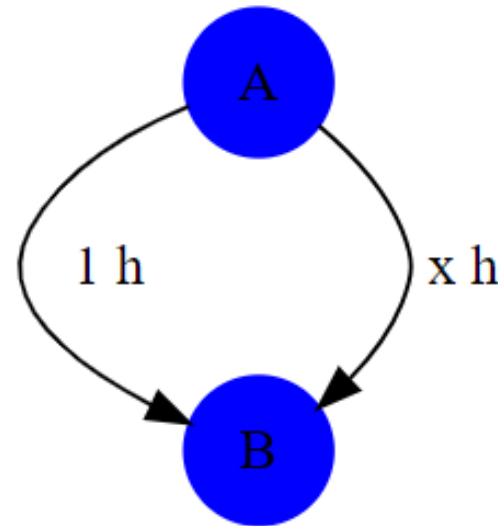
That is 1000 motorists per hour from A to B, two options:

Selfish behavior:

- ▶ Motorists take the path considered at this moment as the fastest empty.
Average time: 1 hour

Social behavior:

- ▶ Each road is closed alternately.
Average time: 45 minutes



Two cities A and B, on either side of a river, connected by two roads.

II. PRESENTATION OF THE ALGORITHM

II.A: Initialization Process

- ▶ **Creation of an arrangement of cities connected by roads according to several parameters**
p: proportions of roads
n: number of cities
- ▶ **Creating car blocks:**
Will leave at time t to get from a city i to a city j by taking the shortest path when traffic is fluid

II. PRESENTATION OF THE ALGORITHM

II.B: Roads

They are characterized by:

The number of lanes, impacting:

1. Their sensitivity to traffic
2. Their **saturation** capacity
3. The maximum **speed** allowed.

The length of the road (bounded in the ALGORITHM between 80 and 200 Km)

1. **Travel time** to travel through it
2. The number of **vehicles** on the road.
3. Their condition: if it is "null", the road is closed.

II. PRESENTATION OF THE ALGORITHM

II.C: The Blocs

They are characterized by:

- ▶ Their **cardinal**: chosen randomly but depending on their period of appearance.
- ▶ The path taken.
- ▶ Their **speed**: constant on the road on which they circulate.
- ▶ Their **status**: "zero" when the block has arrived at its destination or in a queue.

II. PRESENTATION OF THE ALGORITHM

II.D The interface

Ability to iterate manually and see traffic changes, as well as manually close roads.

Afficheur						
Matrice						
-1	0	1	2	3	4	5
0	0.0	1h30	2h42	2h0	0.0	2h42
1	1h30	0.0	1h18	1h12	1h6	1h54
2	2h42	1h18	0.0	1h54	0.0	0.0
3	2h0	1h12	1h54	0.0	1h42	2h0
4	0.0	1h6	0.0	1h42	0.0	0.0
5	2h42	1h54	0.0	2h0	0.0	0.0

Interface

?

Suivant ->

Quitter

Graphe

Journée entière

Courbes

Moyennes

OnOff

0

0

Ouvrir

Fermer

Afficheur TIPE						
Matrice						
-1	0	1	2	3	4	
0	0.0	1.9	0.7	1.0	1.5	
1	4.6	0.0	2.5	4.4	1.8	
2	0.7	2.5	0.0	1.8	1.7	
3	1.0	1.9	2.5	0.0	2.4	
4	1.5	1.7	1.7	1.9	0.0	

Afficheur TIPE						
Matrice						
-1	0	1	2	3	4	
0	0.0	1.7	0.7	1.0	1.5	
1	5.0	0.0	2.5	2.5	2.1	
2	0.7	2.5	0.0	1.8	1.7	
3	1.0	1.7	2.9	0.0	2.3	
4	1.5	1.7	1.7	1.9	0.0	

III. ADVICE FROM THE CONTACT

III.A: Algorithm advice

- ▶ Work with a single graph and then **compare** the results with other arrangements.
- ▶ Is based primarily on **the average of the weights** of the graph.
- ▶ Integrate **a queue system** to avoid outliers.

III. ADVICE FROM THE CONTACT

III.B: Curve proposals

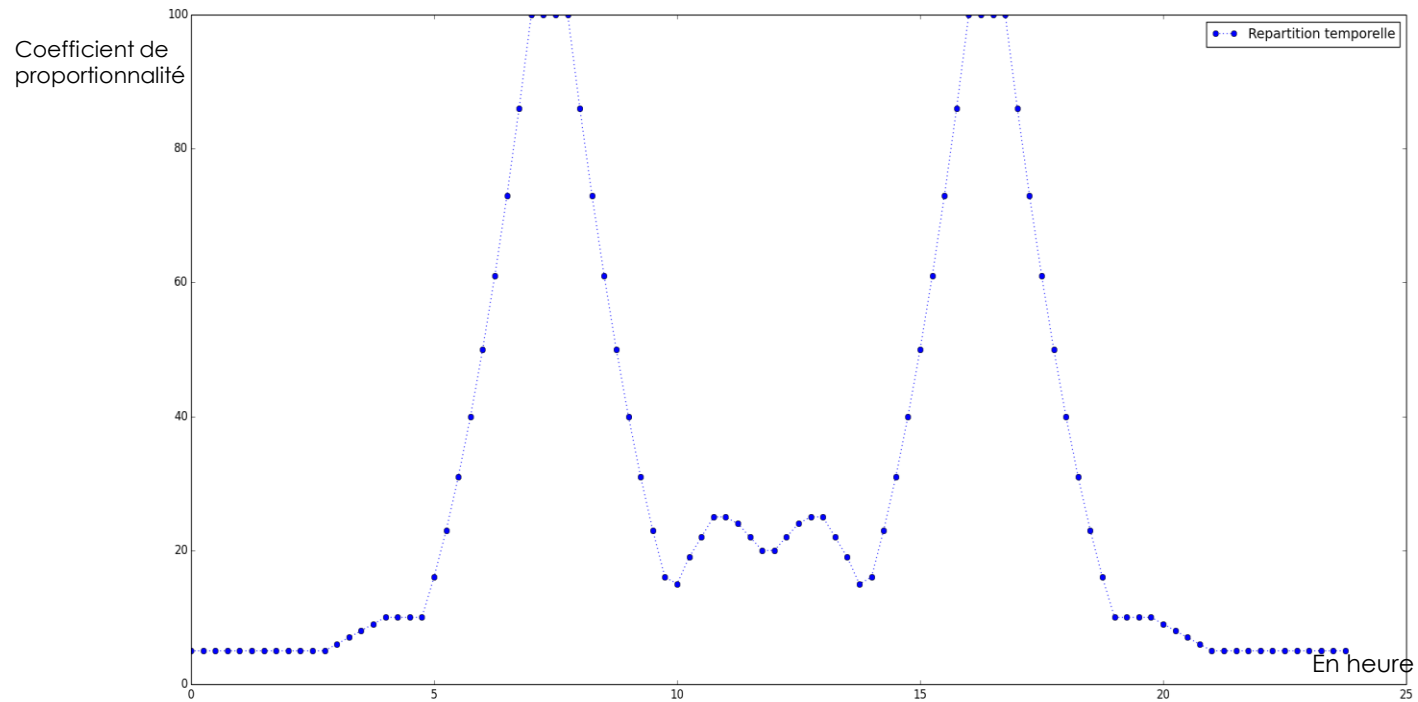
To assess the relevance of the modelling:

- ▶ Compare the evolution of the paradox by varying the **proportion** of roads and the **number** of cities.
- ▶ Compare the evolution of the paradox by varying the closing time of the same road.
- ▶ Evaluate the paradox when closing **widely used** roads compared to others that are not very used.

Result: our curves are quite plausible, the modelling is **admissible.**

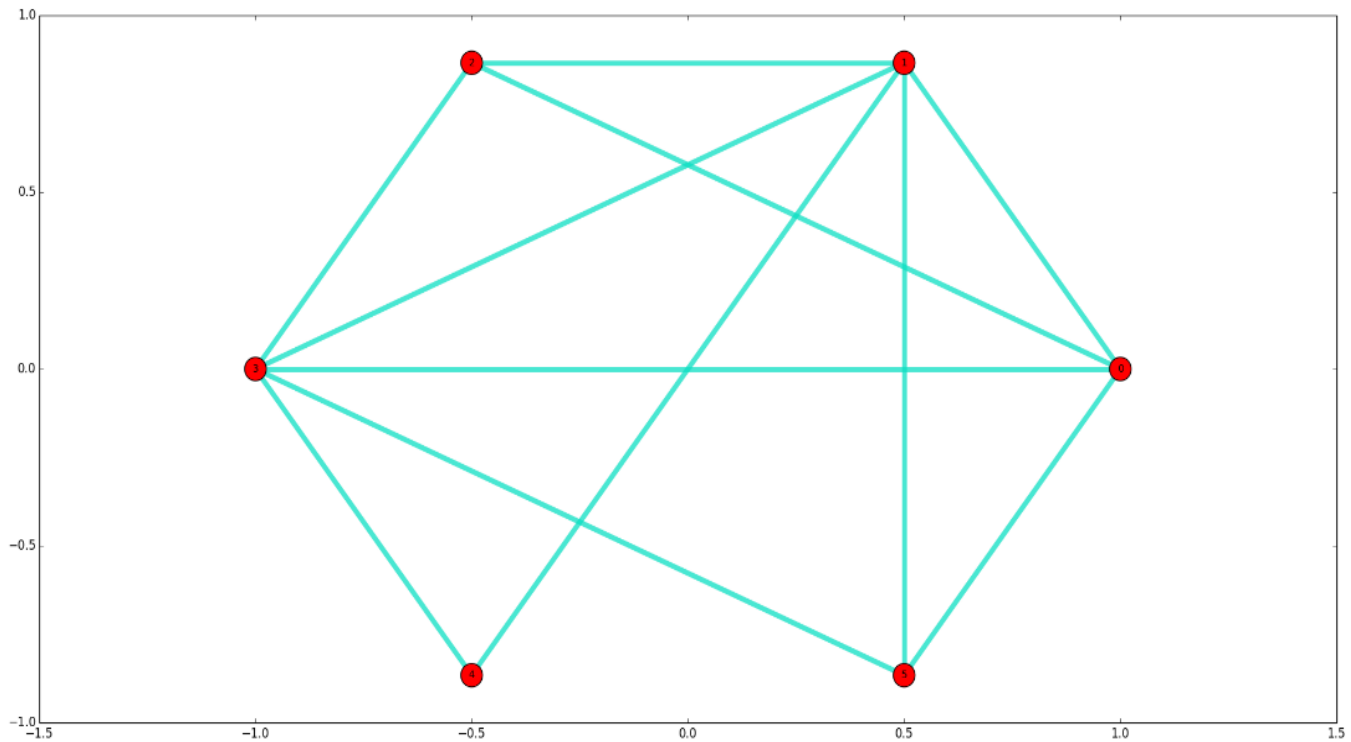
IV TRAFFIC ANALYSIS

IV.A Temporal distribution



IV TRAFFIC ANALYSIS

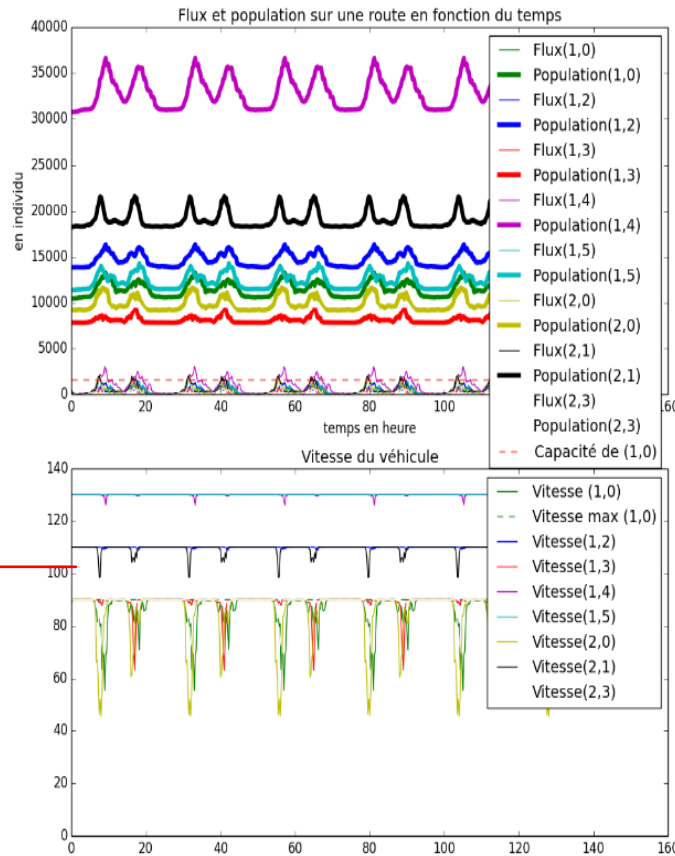
IV.B Traffic study for 6 cities



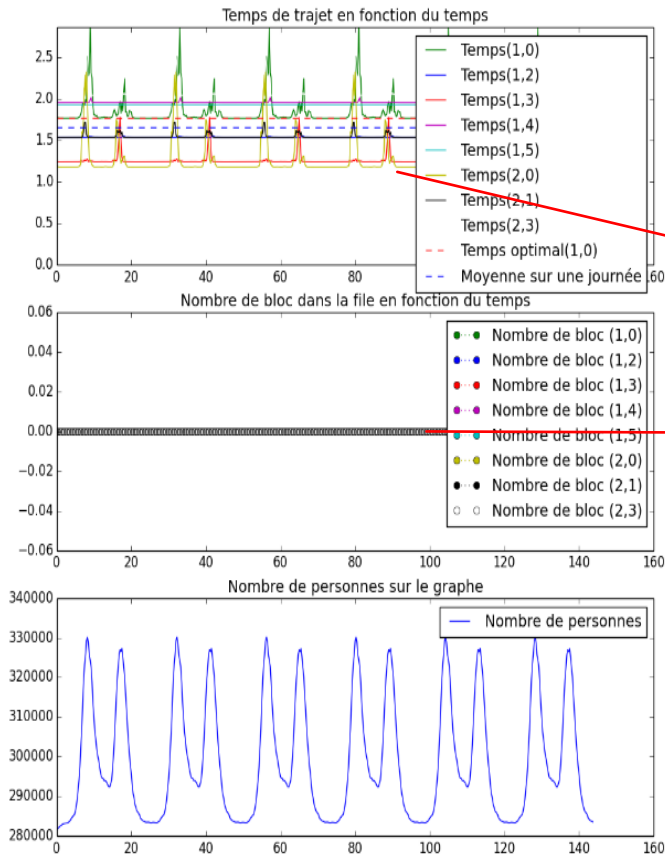
IV TRAFFIC ANALYSIS

IV.B Traffic study for 6 cities

Little Headcount



Correlation
between
curves



Little slowdown

No blocks
queued

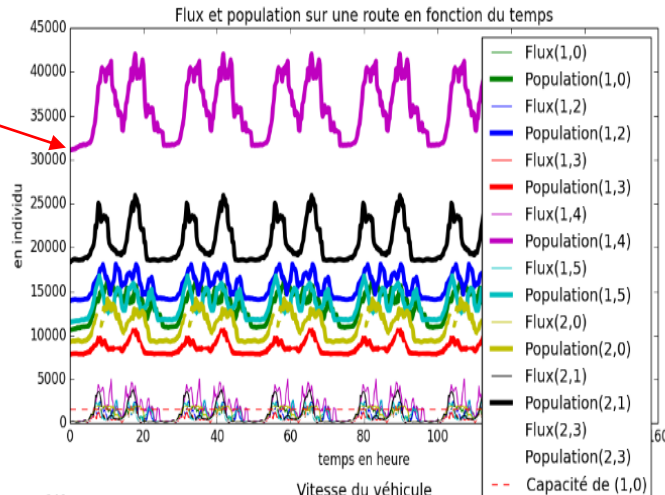
Average over a day of 1.6 hours with a critical population of 330000.

IV TRAFFIC ANALYSIS

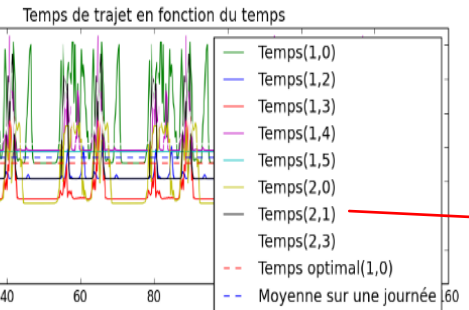
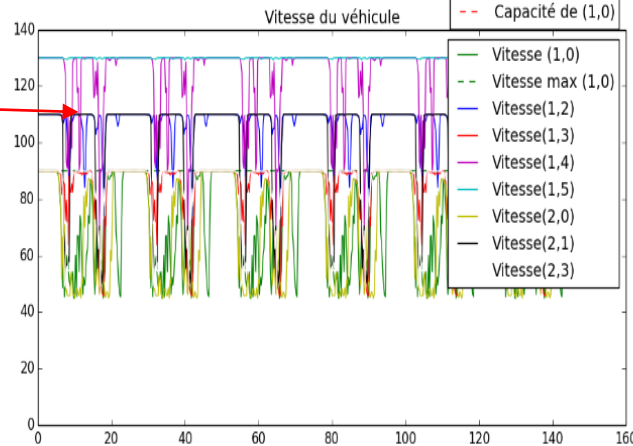
IV.B Traffic study for 6 cities

Average headcount

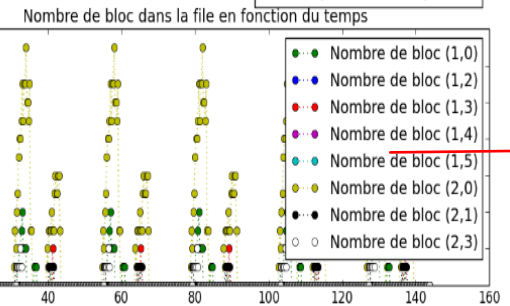
Prioritization



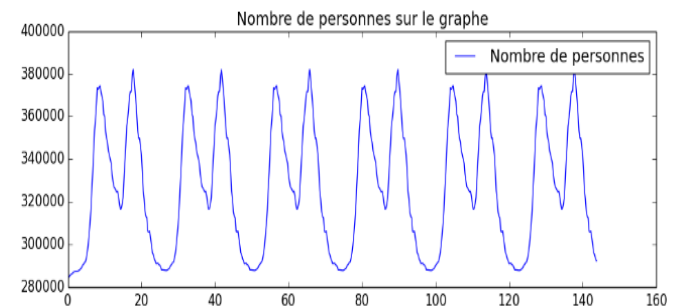
The cap spreads on the yellow curve.



The peaks follow the same pace by amplifying



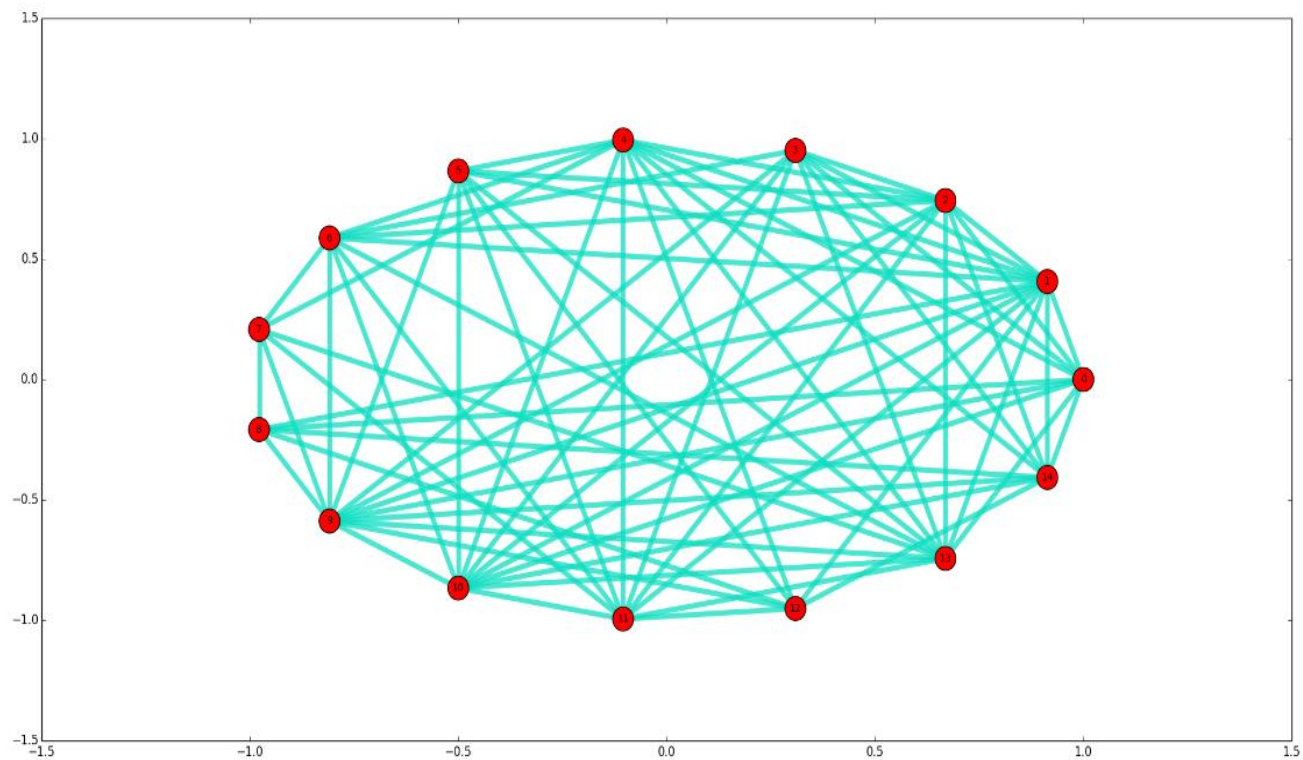
The yellow curve has a queue.



Average over a day of 1.8 hours with a critical population of 380,000.

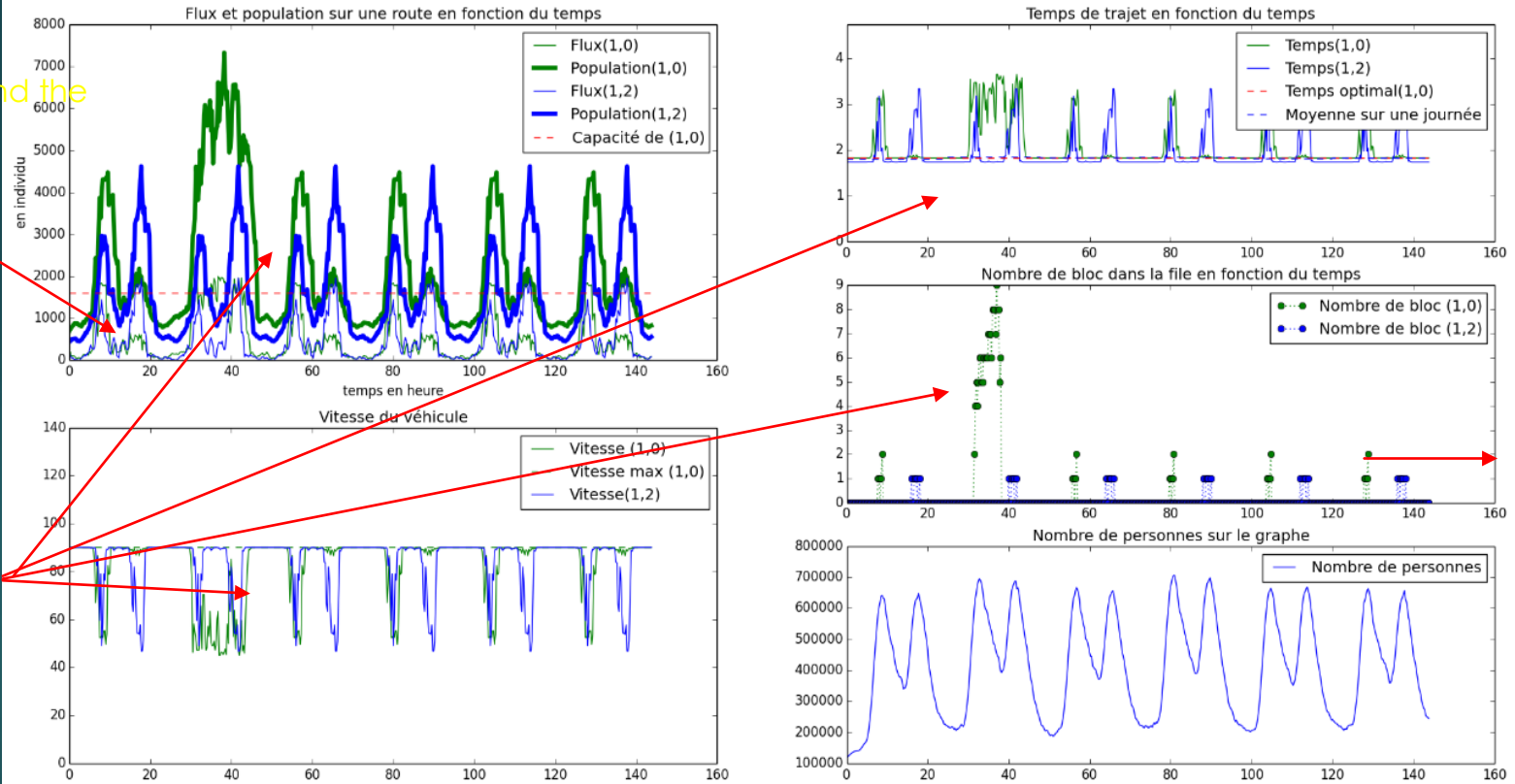
IV TRAFFIC ANALYSIS

IV.C Traffic study for 15 cities



IV TRAFFIC ANALYSIS

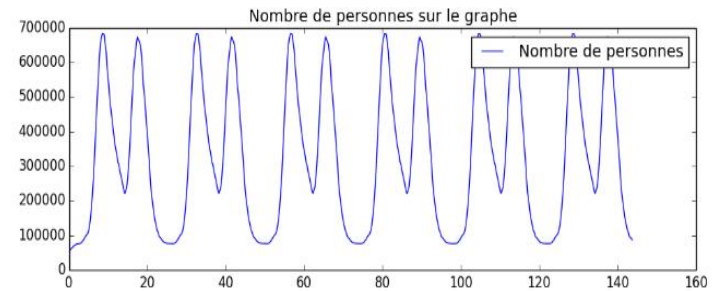
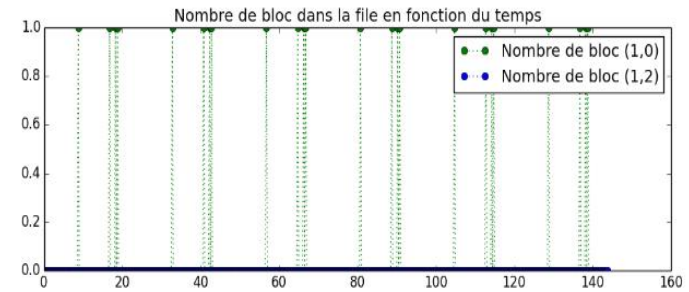
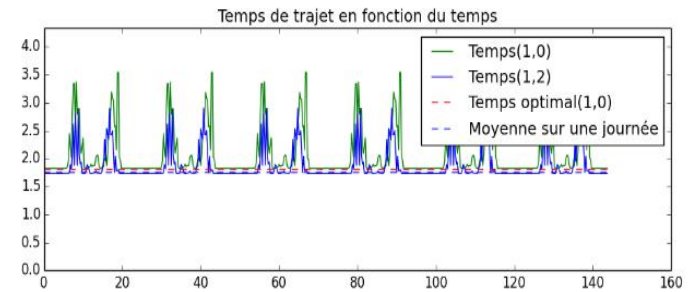
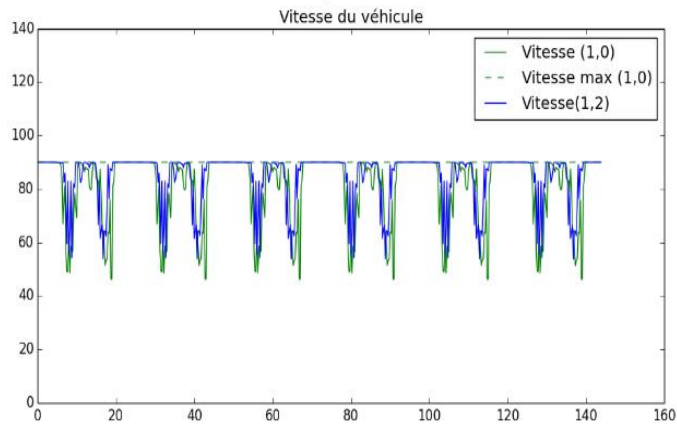
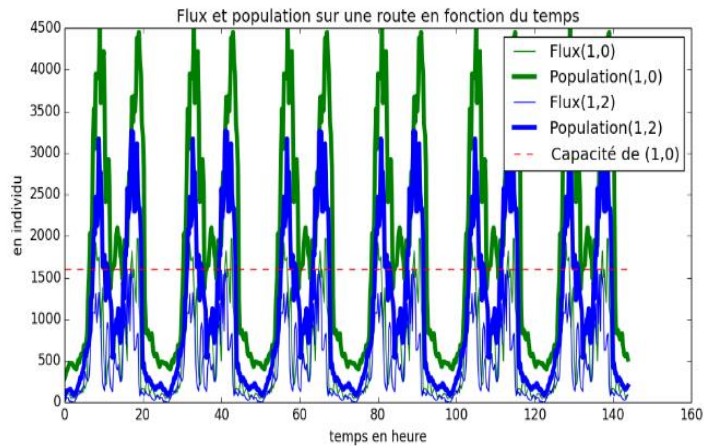
IV.C Traffic study for 15 cities



Average over a day of **1.7 hours** with a critical population of **600,000**.

IV TRAFFIC ANALYSIS

IV.C Traffic study for 15 cities(Social)



There is a disappearance of traffic jams

IV TRAFFIC ANALYSIS

IV.D: Approach to the price of anarchy

Definition 1: Routing is a possible configuration of traffic.

- Theorem 1: There is **social** routing.
- Theorem 2: There is **selfish** routing. By Beckmann, McGuire and Winsten (1956)

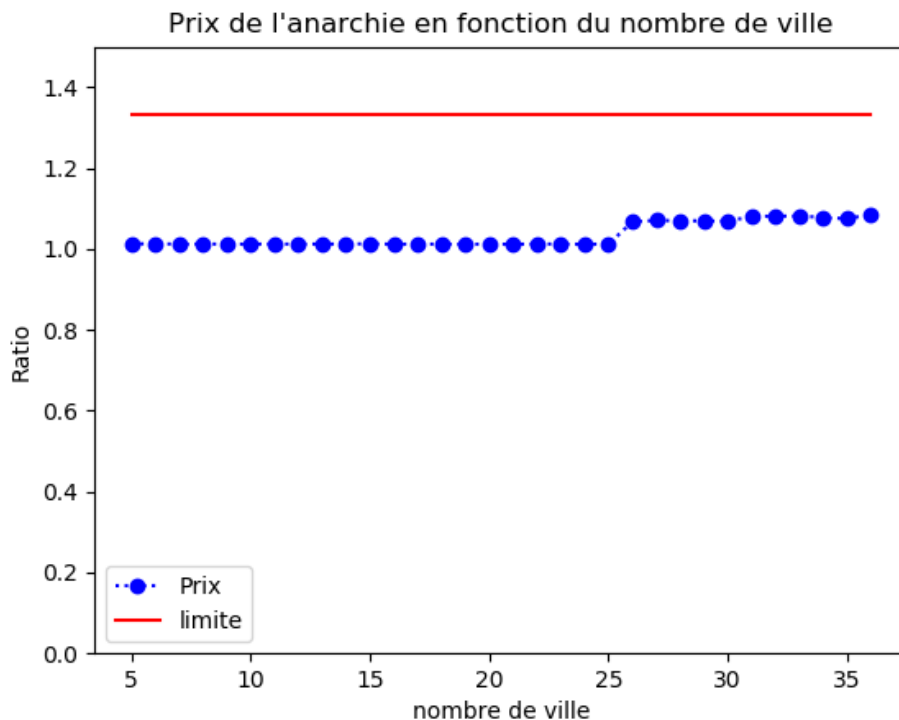
Definition 2 : The price of anarchy is defined as the ratio of the average time of selfish routing to the average time of social routing.

Théorème 3 : The price of anarchy does not exceed **4/3**. By Roughgarden and Tardos (2002)

IV TRAFFIC ANALYSIS

IV.D: Approach to the price of anarchy

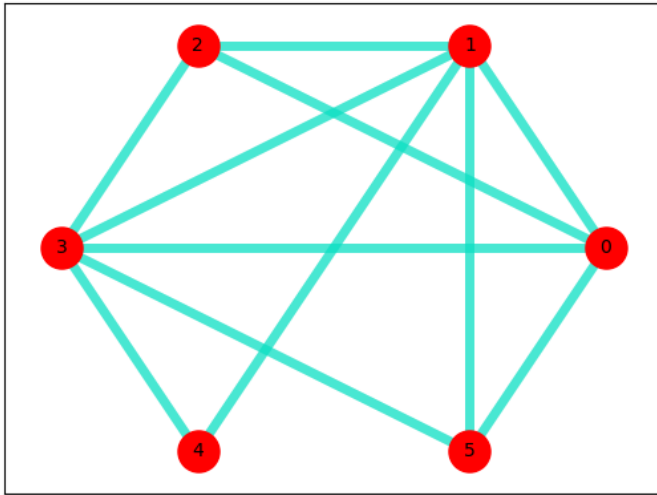
What our algorithm thinks



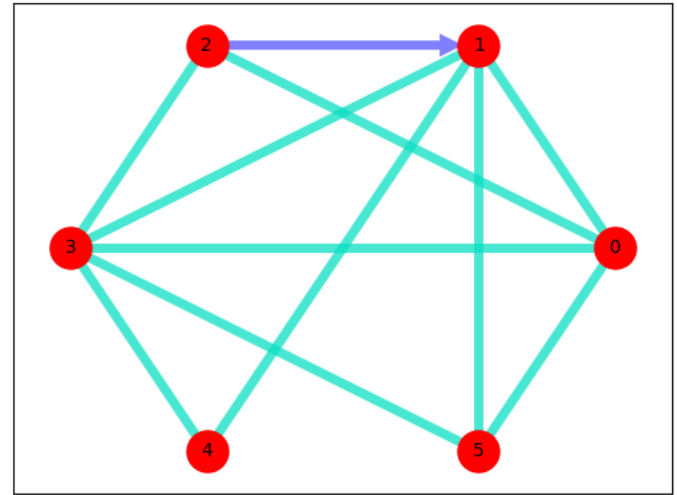
These are results that seem consistent, social routing is **lower and weaker** when n increases relative to selfish routing, which is likely to **increase** when there are a larger number of people on the graph.

V. STUDY OF THE PARADOX

V.A By closing on a one-hour time slot for 6 Cities:



Before



After

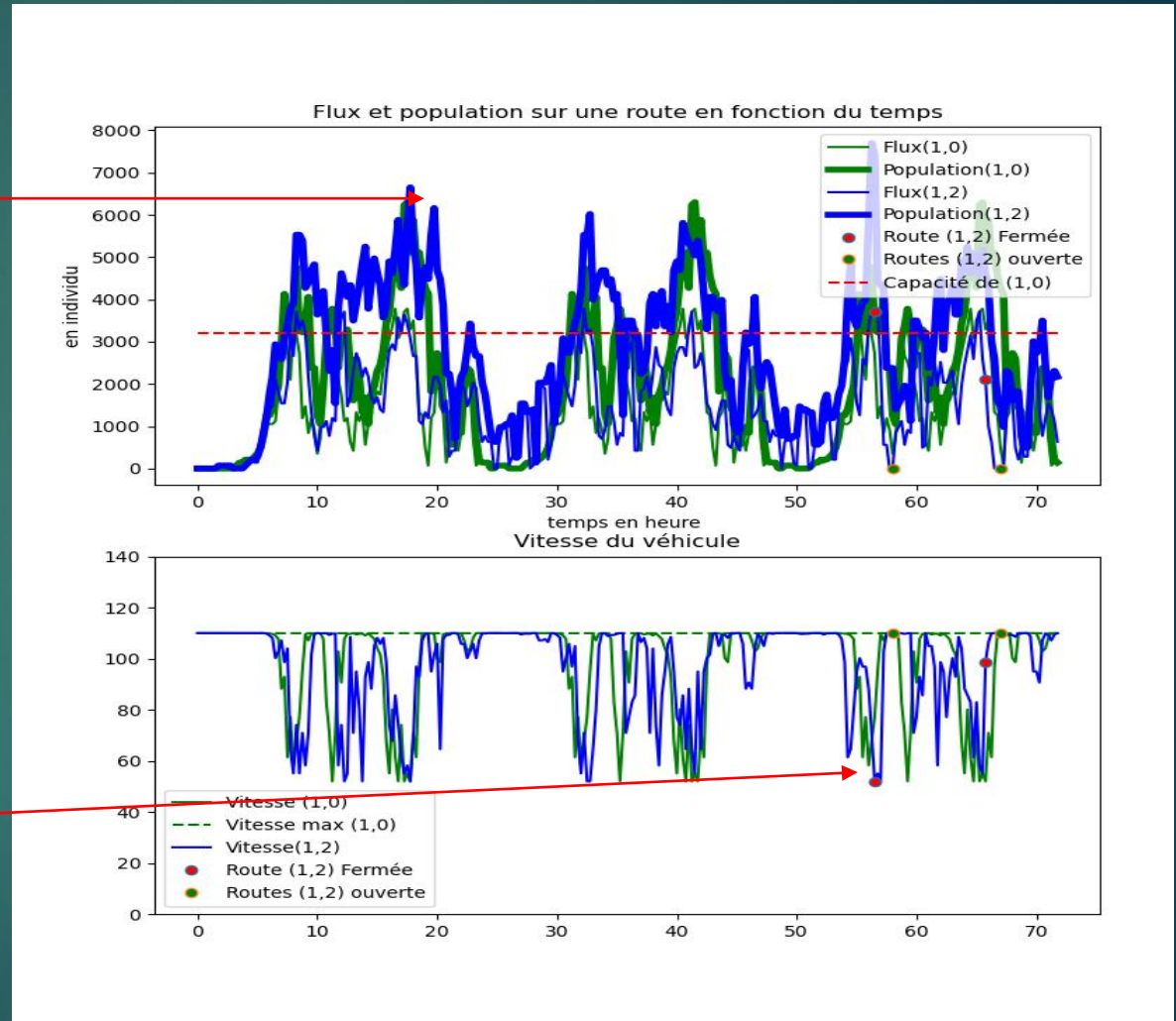
V. STUDY OF THE PARADOX

V.A.1 Situation on the road and other roads – Comparison of the last day of simulation against the first two

Here is the most used road during busy periods that must be closed (1.2 here)

The flow after closing decreases and the usual speed is restored on the closed road.

Closing and opening on the third day.



Roads need to be clogged for this to work

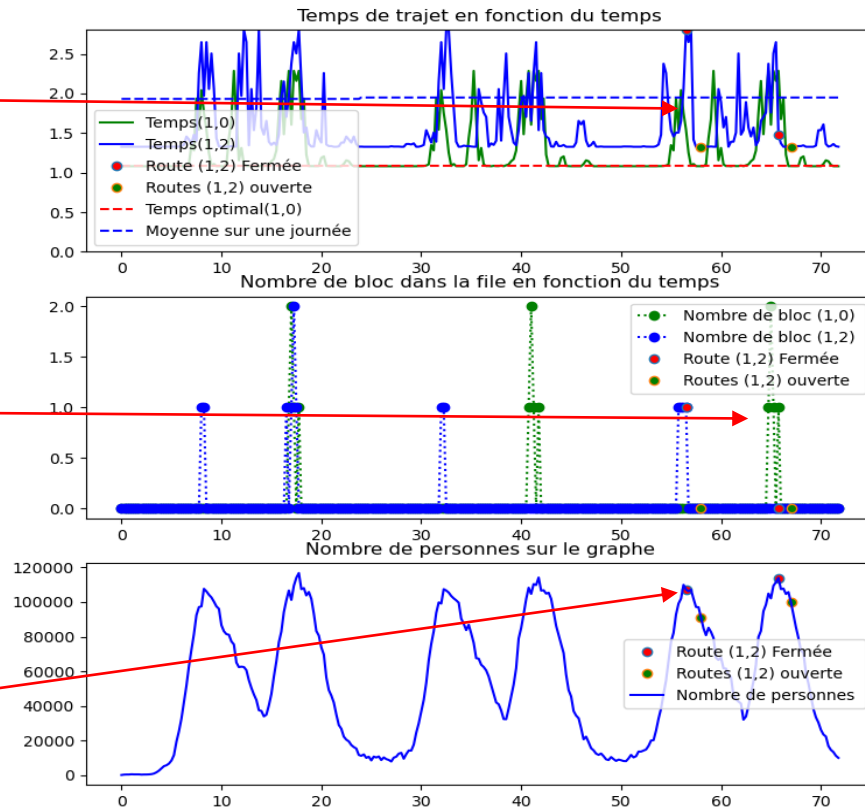
V. STUDY OF THE PARADOX

V.A.2 Situation on the road and other roads – Comparison of the last day against the first two

The green road then becomes more used than usual, but is not saturated!

There is no longer a traffic jam (block waiting) on the blue road.

Slight decrease in population



We understand that by decreasing the flow on one road, we increase it on another. This would have an advantage when the closed road is saturated.

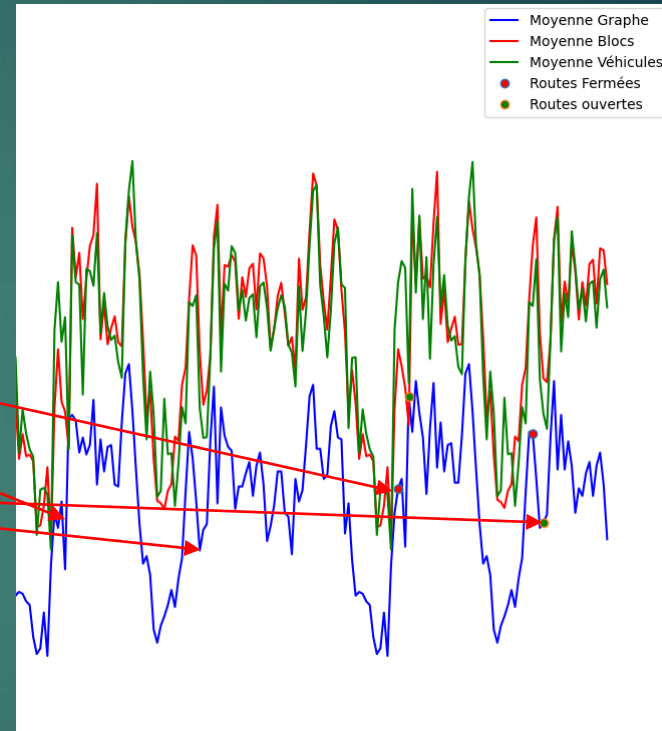
V. STUDY OF THE PARADOX

V.B Effect on the whole graph – comparison between a day without closure and with closure.

The average time during peaks is not necessarily reduced

There are fewer irregularities over the closing period

However, the average time following the peak period is slightly higher.



Temps moyen

Deux jours consécutifs

VI: Conclusion

- ▶ The curves modeling a circulation offered by our algorithm are admissible which confirms the relevance of our program.
- ▶ The tests, particularly with regard to closures on a time slot, remain **relative** because they are based only **on a single arrangement**, however it emerges:
 - For roads in the vicinity:
 - These will be more borrowed.
 - The clogged road will open.
 - This avoids saturating a single axis.
 - For everyone:
 - Less irregularity in traffic
 - However, we must not be mistaken in the road and moment.
- ▶ Finally it is necessary to close the **most used roads** and **do it during high traffic**

VI. Conclusion

- ▶ The price of anarchy is verified experimentally
- ▶ The case of social routing would be a perfect optimization of the closures of each road.
- ▶ In this case, the irregularities disappear and the system becomes perfectly stable.
- ▶ Travel time is completely optimized for all users.
- ▶ Some users, however, will go slower than if he had been selfish.

VII. ANNEXE

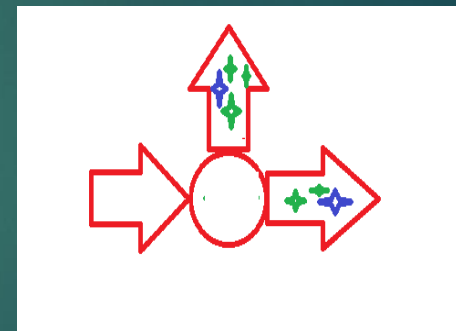
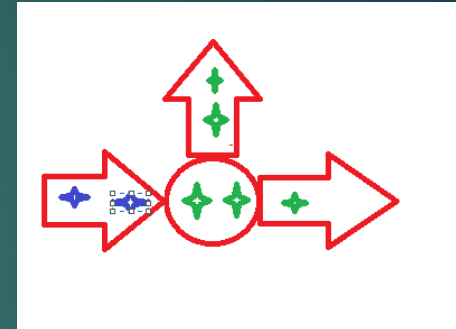
VII.A Iteration Process

Inserting blocks:

Of those who appear on the first road they will take.

- ▶ Of those who made it to the end of the road in the previous iteration on their next route

- ➡ Addition of the population of these blocks to the incoming flow.
- ➡ Assignment of a speed v which depends on the current weight of the road (permanent on it).



VII. ANNEXE

VII.A Iteration Process

Evaluation of the other blocks:

- ▶ If it is likely to modify the **flow** so that it exceeds **the saturation threshold too much**.
 - ➡ It is placed in the **queue** corresponding to the route.
- ▶ Otherwise we assign him his new **position** by considering that he has moved at this speed **v**.

VII. ANNEXE

VII.A Iteration Process

Road evaluation:

- ▶ Knowing the incoming **flow** over an hour, they are assigned their new **weight** (travel time).
 - ▶ Refreshing the **flow**
 - ▶ Evaluation of our different **queues** :
-  If adding a block does not cause too high a flow, the block starts from the queue and enters the road.

VII. ANNEXE

VI.B Road capacity

7 Capacité pratique de différents types de routes

Ces valeurs sont applicables à des tronçons homogène de routes ou autoroutes

- En rase campagne
- En terrain plat
- Sans carrefours
- Avec une répartition du trafic 60/40 pour les voies bidirectionnel

Type de la route	Seuil de gène	Seuil de saturation
2 voies	8500	15000
3 Voies	12000	20000
2 x 2 Voies	25000	45000
2 x 3 Voies	40000	65000

8 Capacité des routes (B40)

Capacité des routes en UVP/ Heure									
Suivant la largeur de la chaussée et les dégagements latéraux									
Dégagement latéral ou largeur des accotements (des 2 cotes de la chaussée)	Largeur de chaussée unique								Deux (2) chaussées
	2 voies				3 Voies		4Voies		Autoroutes et Routes express
	4 m	5m	6m	7 m	9m	10.5m	12 m	14 m	
1,80	1100	1300	1600	2000	2600	3200	3000 (4500)	3400 (5100)	1800 (5400)
1,20	1000	1200	1500	1900	2400	3000	2950 (4400)	3300 (5000)	1750 (5300)
0,60	X	1100	1350	1700	2200	2700	2900 (4300)	3200 (4800)	1700 (5100)
0,00	X	X	1200	1500	2000	2400	2650 (4000)	3000 (4500)	1500 (4500)

VII.C The Law of

$$\tau_V(x_V, x_C) = Lm_V \left[1 + \gamma_V \left(\frac{x_V + ex_C}{k} \right)^{\alpha_V} \right] \text{ avec :}$$

τ : temps de parcours VP

L : longueur de la relation considérée

x_V, x_C : débit horaire des VL, des PL

$x_V + ex_C$: débit horaire total en uvp

m_V : temps de parcours moyen kilométrique des VP, à vide

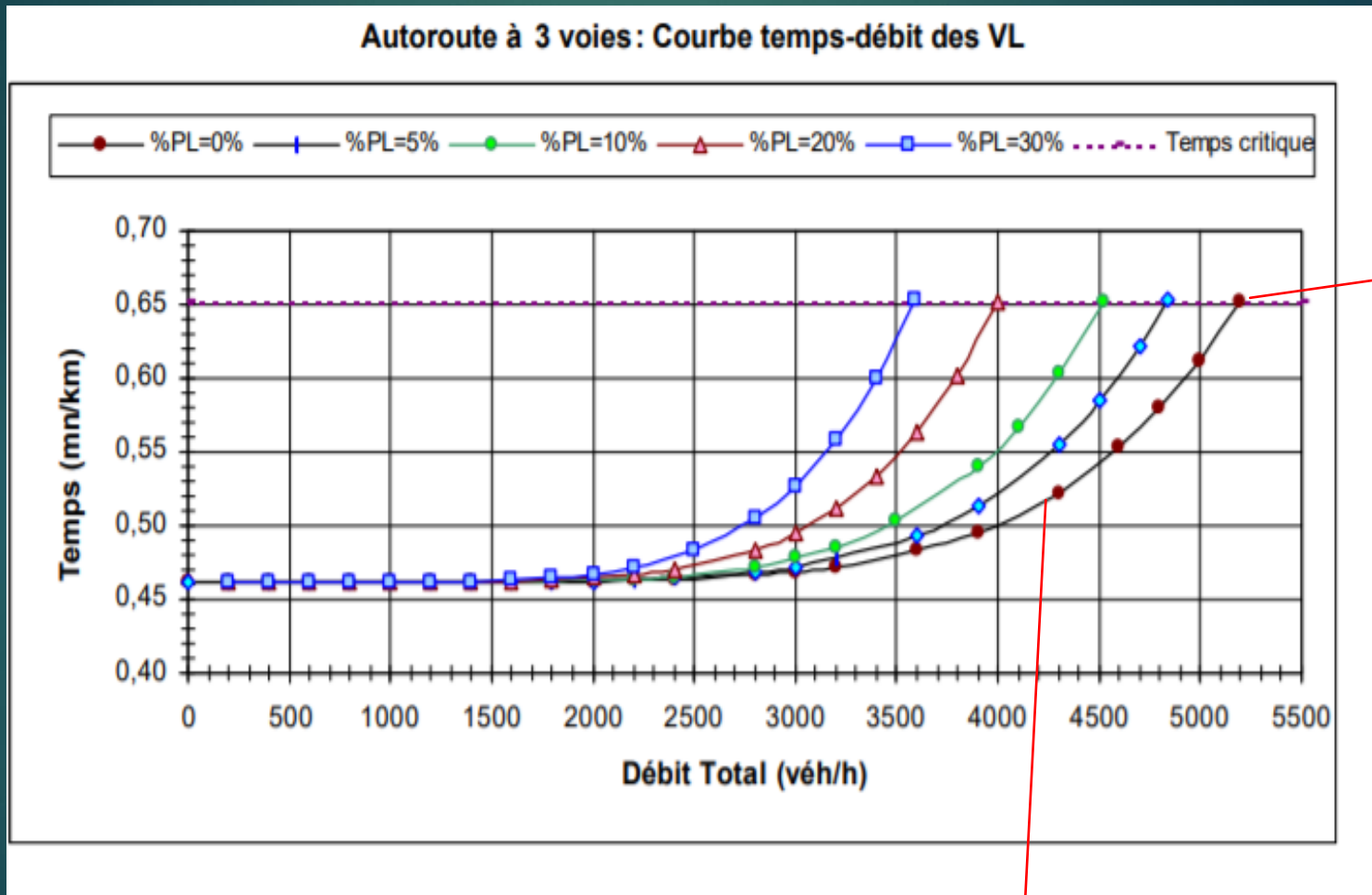
γ_V : proportion d'augmentation du temps à la saturation. Par exemple si $a = 0,15$ cela signifie que le temps de parcours est supérieur de 15 % à celui dit « à vide ».

α_V : sensibilité du temps de parcours au débit relatif $(x_V + ex_C)/k$ à l'approche de la saturation

VII. ANNEXE

VII.C The Law of Speed

An overview



Seuil de gêne

VII ANNEXE

VII.D: Presentation

Résumé

Recherche Opérationnelle :

- Heuristiques spécifiques et adaptation d'algorithmes classiques
- Recherche de plus court chemin sous contraintes
- Problèmes de tournées
- Optimisation de chargements
- Programmation mathématique

Statistiques :

- Big Data
- Économétrie
- Prévisions
- Analyse statistique de données
- Segmentation

Intelligence artificielle

- Machine Learning
- Réseau de neurones
- Propagation de contraintes

Modélisation probabiliste

- Réseau bayésien
- Processus de Markov (files d'attente)



Thierry Heinis

Scientific Expert DATA Scientist

Région de Paris, France

Technologies et services de l'information

 EURODECISION

 Ecole Centrale Paris

 Université Paris Dauphine

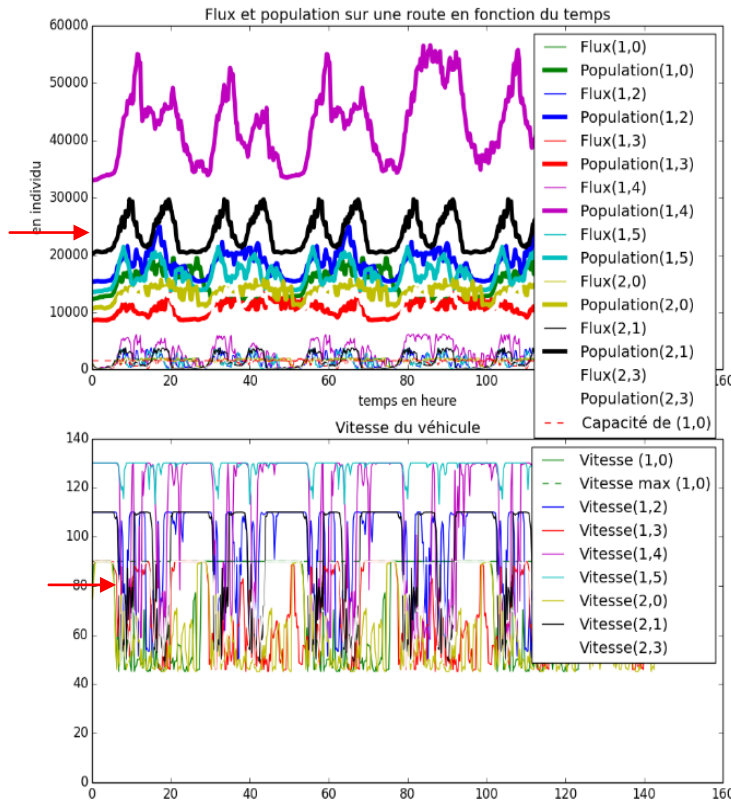
 Site web de l'entreprise

+ de 500 relations

VII Annexe

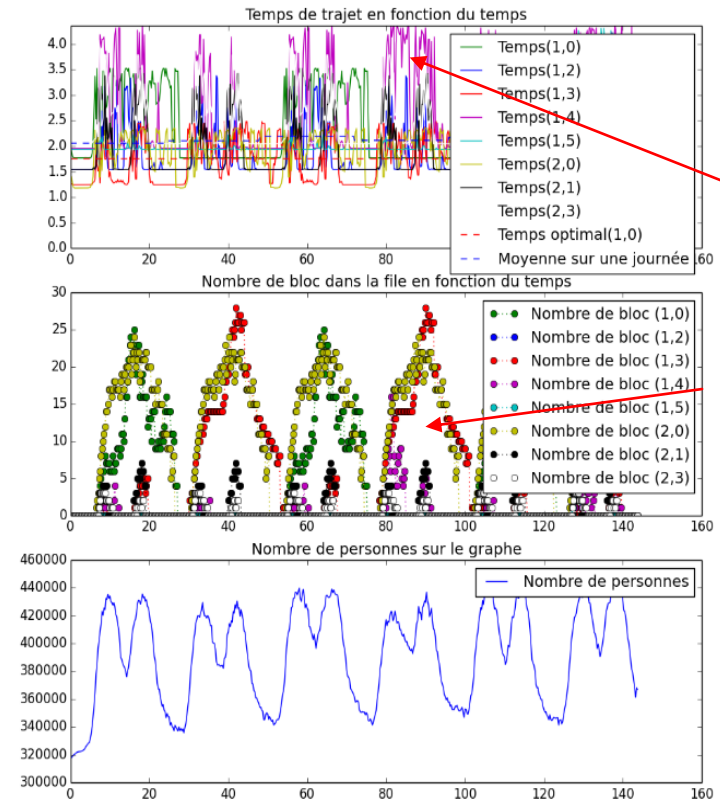
VII.E: Effect of social routing on a large workforce

Large population



Clogged roads are always clogged

Speeds aren't stable



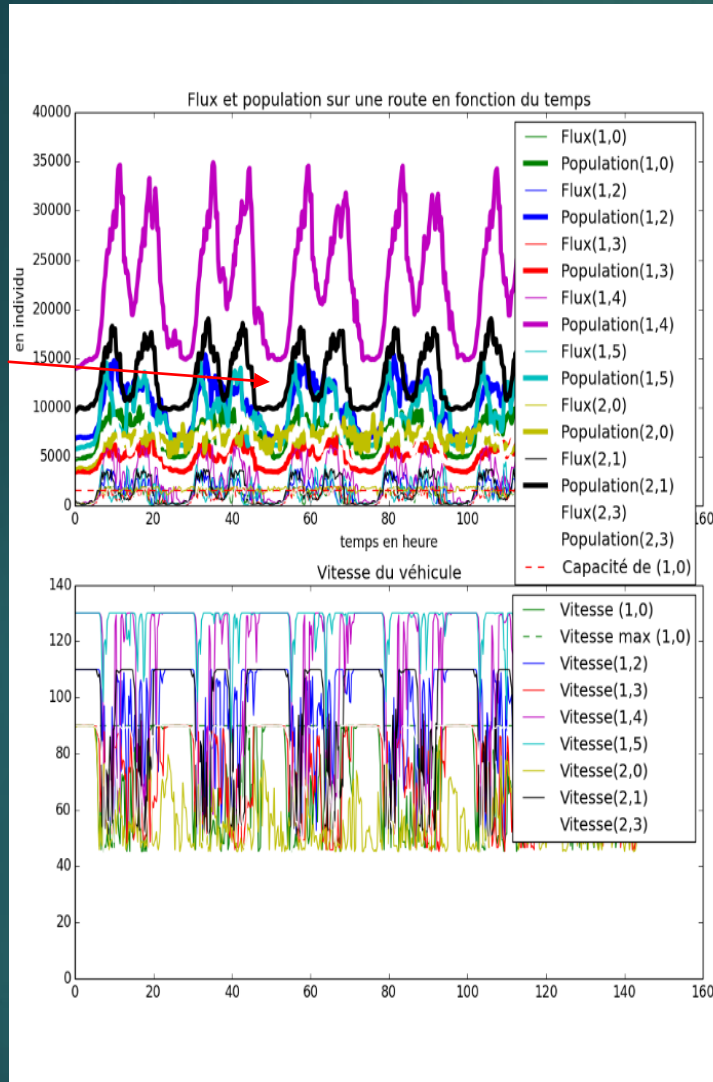
The purple curve is saturated

The red curve in turn clogs

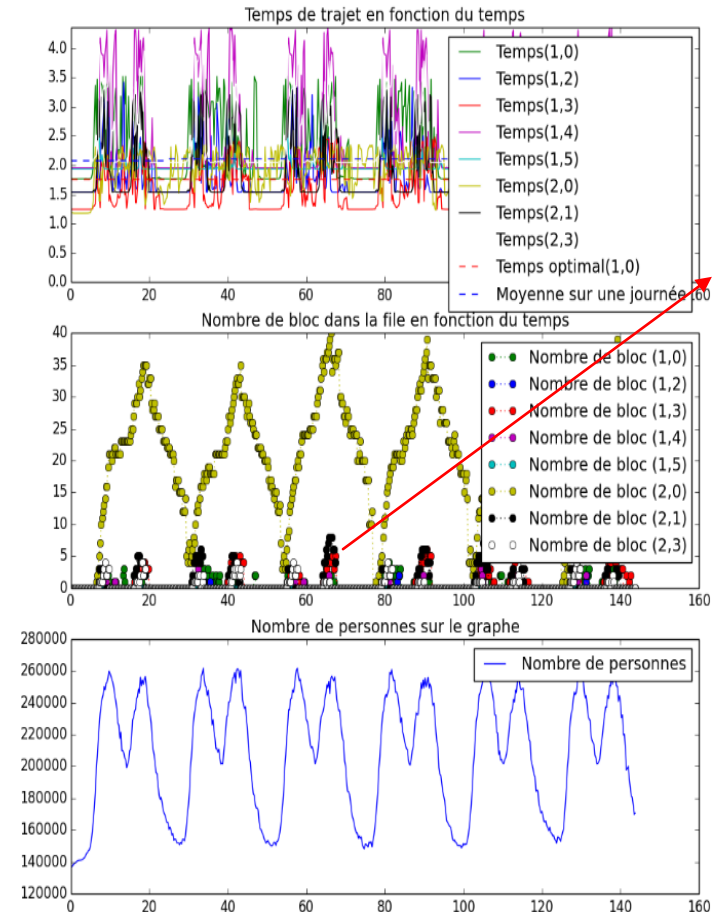
Average over a day of 2.1 hours with a critical workforce of 440000

VII Annexe

VII.E Effect of social routing on a very large workforce



The plugs no longer spread, the peaks are more hollow.



The red curve is no longer clogged.

Average over a day of 1.8 hours with a critical workforce of 260000